

Problem 8– Flexural Strength

The cross-section of a post-tensioned concrete beam spanning 16 m, simply supported, is shown in Figure 8.1. The beam is post-tensioned with a tendon, which has a parabolic profile, with end eccentricity zero and maximum eccentricity at midspan of 350 mm. The tendon comprises of 10 number 12.7 mm super grade low relaxation strands. Tendon is stressed to 70% of the breaking load. Assume immediate losses 15% and deferred losses 10%

Properties of materials are:

Concrete:

$$f'_c = 32 \text{ MPa}$$

$$f'_{cp} = 25 \text{ MPa (at 14 days when the tendon is stressed)}$$

Prestressing Strands:

$$\text{Cross-sectional area of a single strand} = 100 \text{ mm}^2$$

$$\text{Yield strength of tendons, } f_{py} = 1640 \text{ MPa}$$

$$\text{Minimum breaking load of strands} = 184 \text{ kN}$$

$$E_p = 190,000 \text{ MPa}$$

$$d_p = 850 \text{ mm}$$

Ordinary Reinforcement:

$$A_{st} = 8N24 = 3600 \text{ mm}^2, d_{st} = 963 \text{ mm}$$

$$E_p = 190,000 \text{ MPa}, E_s = 200,000 \text{ MPa}$$

1. Calculate the moment capacity of the sections at ultimate moment strength
2. Check whether the section would be safe at transfer.

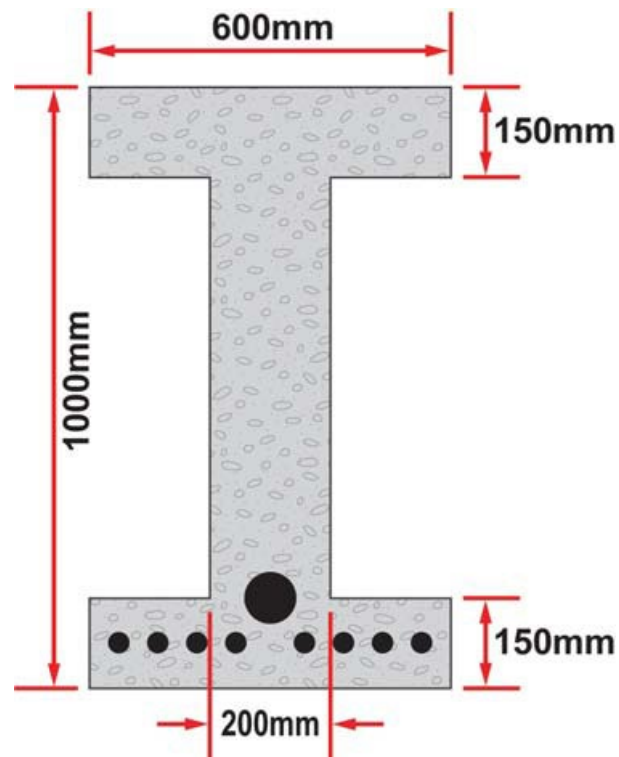


Figure 8.1 – Post-tensioned beam section

Problem 9– Shear Strength

The cross-section of a post-tensioned concrete beam spanning 16 m, simply supported, is shown in Figure 9.1. The beam is post-tensioned with a tendon, which has a parabolic profile, with end eccentricity zero and maximum eccentricity at midspan of 350 mm. The tendon comprises of 10 number 12.7 mm super grade low relaxation strands. Tendon is stressed to 80% of the breaking load. Assume immediate losses 5% and deferred losses 20%. Ignore friction loss. Properties of materials are:

Live load = 14kN/m, 112mm duct

Concrete:

$$f'_c = 32 \text{ MPa}$$

$$f'_{cp} = 25 \text{ MPa (at 14 days when the tendon is stressed)}$$

Prestressing Strands:

$$\text{Cross-sectional area of a single strand} = 100 \text{ mm}^2$$

$$\text{Yield strength of tendons, } f_{py} = 1640 \text{ MPa}$$

$$\text{Minimum breaking load of strands} = 184 \text{ kN}$$

$$E_p = 190,000 \text{ MPa}$$

$$d_p = 850 \text{ mm}$$

Ordinary Reinforcement:

$$A_{st} = 8N24 = 3600 \text{ mm}^2, \quad d_{st} = 950 \text{ mm}$$

$$E_p = 190,000 \text{ MPa}, \quad E_s = 200,000 \text{ MPa}$$

1. Design a critical section near the support for shear
2. Design for shear at quarter span ($x=4\text{m}$)

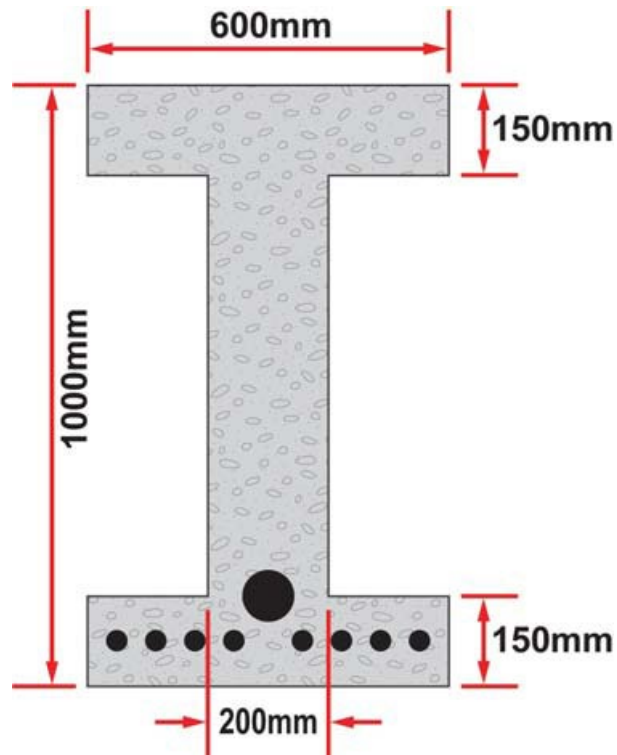


Figure 9.1 – Post-tensioned beam section